Chapter 9. Fourier Series

Orthonormal systems, pointwise convergence, Fejér's theorem, Parseval's identity, smoothness and decay

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9.1 Fourier coefficients

Motivation. Fourier coefficients give the best L^2 approximation by trigonometric polynomials.

In an inner product space X, a family $\{\phi_n : n \in \mathbb{Z}\}$ is **orthonormal** if $\langle \phi_m, \phi_n \rangle = \delta_{mn}$. In $\mathcal{R}^2[-\pi,\pi]$ with $\langle f, g\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} f\bar{g}$, the functions $e_n(x) = e^{inx}$ are orthonormal. For $f \in \mathcal{R}^1[-\pi,\pi]$ (extended 2π-periodically) the **Fourier coefficients** are $\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-int}dt$, the **Fourier series** is $\sum \hat{f}(n)e^{inx}$, and $s_n(f;x) = \sum_{|k|\le n}\hat{f}(k)e^{ikx}$.

Theorem 9.1 (9.1.1, best approximation and Bessel's inequality)

Let $\{\phi_n\}$ be orthonormal in X, $x \in X$, $c_n = \langle x, \phi_n\rangle$. For all complex z_n,

$$\Big\|x - \sum_{n=-N}^{N} c_n\phi_n\Big\| \le \Big\|x - \sum_{n=-N}^{N} z_n\phi_n\Big\|,$$

with equality iff $z_n = c_n$ for all $|n| \le N$. Moreover $\sum_{n\in\mathbb{Z}}|c_n|^2 \le \|x\|^2$.

Proof. Let $s = \sum_{|n|\le N} c_n\phi_n$ and $t = \sum z_n\phi_n$. Then $\langle x-s, \phi_n\rangle = 0$ for $|n| \le N$, so $x - s \perp s - t$, and $\|x-t\|^2 = \|x-s\|^2 + \|s-t\|^2 = \|x-s\|^2 + \sum|c_n - z_n|^2$. With $t = 0$: $\|x\|^2 = \|x-s\|^2 + \sum_{|n|\le N}|c_n|^2$, which gives Bessel's inequality. ∎

Proposition 9.2 (9.1.2)

For $f \in \mathcal{R}^1[-\pi,\pi]$:
(a) if f is real, $\hat{f}(-n) = \overline{\hat{f}(n)}$;
(b) if f is even, $\hat{f}(-n) = \hat{f}(n)$;
(c) if f is odd, $\hat{f}(-n) = -\hat{f}(n)$.

Proof. (a) Conjugate the integral. (b), (c) Substitute $t \mapsto -t$. ∎

In real form,

$$s_n(f;x) = \frac{a_0}{2} + \sum_{k\le n}(a_k\cos kx + b_k \sin kx)$$

with $a_k = \frac{1}{\pi}\int f(t)\cos kt\,dt$, $b_k = \frac{1}{\pi}\int f(t)\sin kt\,dt$; even functions have cosine series and odd functions sine series.

Example 9.1

$f(x) = x$ on $(-\pi,\pi)$: $\hat{f}(n) = \frac{(-1)^n i}{n}$ $(n \ne 0)$, so $x \sim 2\sum(-1)^{n+1}\frac{\sin nx}{n}$. $f(x) = |x|$: $|x| \sim \frac{\pi}{2} - \frac{4}{\pi}\sum_{k \text{ odd}}\frac{\cos kx}{k^2}$, and at $x = 0$ this gives $\sum_{k\text{ odd}} k^{-2} = \pi^2/8$, hence $\sum n^{-2} = \pi^2/6$ (once convergence to f is proved, Corollary 9.7).

9.2 Pointwise convergence

Motivation. The Fourier series converges to f where f is reasonably smooth: Dirichlet kernel plus Riemann–Lebesgue.

Theorem 9.3 (9.2.1, Riemann–Lebesgue lemma)

If $\int_I |f| < \infty$ on an interval I, then for every $\beta \in \mathbb{R}$, $\lim_{\alpha\to\infty} \int_I f(t)\sin(\alpha t+\beta)\,dt = 0$.

Proof. For $f = \chi_{[a,b]}$, $|\int_a^b \sin(\alpha t+\beta)dt| \le 2/\alpha$; by linearity the claim holds for step functions. For general f and ϵ, choose a step function s with $\int_I |f-s| < \epsilon$ (Theorem 7.16); then $|\int f\sin(\alpha t+\beta)| \le \epsilon + |\int s\sin(\alpha t+\beta)| < 2\epsilon$ for large α. ■

In particular $\hat{f}(n) \to 0$ as $|n| \to \infty$ (Theorem 9.21). The partial sums are integrals against the **Dirichlet kernel**:

$$s_n(f;x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)D_n(t)\,dt, \qquad D_n(t) = \sum_{|k|\le n} e^{ikt} = \frac{\sin(n+\frac{1}{2})t}{\sin\frac{t}{2}}, \qquad \frac{1}{2\pi}\int_{-\pi}^{\pi} D_n = 1.$$

Since D_n is even, $s_n(f;x) = \frac{1}{2\pi}\int_0^{\pi}[f(x+t)+f(x-t)]D_n(t)\,dt$.

Theorem 9.4 (9.2.2, Riemann localization)

Let $0 < \delta < \pi$ and $\int_{-\pi}^{\pi}|f| < \infty$. Then $\lim_n s_n(f;x)$ exists iff $\lim_n \frac{1}{\pi}\int_0^{\delta}[f(x+t)+f(x-t)]\frac{\sin(n+\frac{1}{2})t}{t}\,dt$ exists, and then they are equal. So convergence of the Fourier series at x depends only on f near x.

Proof. $\frac{1}{\sin(t/2)} - \frac{2}{t}$ is bounded on $(0,\pi]$, so by Theorem 9.3, $\frac{1}{2\pi}\int_0^{\pi}[f(x+t)+f(x-t)](\frac{1}{\sin(t/2)} - \frac{2}{t})\sin(n+\frac{1}{2})t\,dt \to 0$; and $\frac{2}{t}$ is bounded on $[\delta,\pi]$, so the integral over $[\delta,\pi]$ with kernel $\frac{2\sin(n+\frac{1}{2})t}{t}$ also tends to 0. ■

We use $\lim_{\lambda\to\infty}\int_0^{\delta}\frac{\sin\lambda t}{t}\,dt = \lim\int_0^{\lambda\delta}\frac{\sin u}{u}du = \frac{\pi}{2}$ (Example 8.2).

Lemma 9.5 (9.2.3)

If $g(0+)$ exists and $\int_0^{\delta}\frac{|g(t)-g(0+)|}{t}\,dt < \infty$, then $\lim_{\lambda\to\infty}\frac{2}{\pi}\int_0^{\delta} g(t)\frac{\sin\lambda t}{t}\,dt = g(0+)$.

Proof. Write $g(t) = g(0+) + (g(t)-g(0+))$. The first part gives $g(0+)\cdot\frac{2}{\pi}\cdot\frac{\pi}{2}$ in the limit; the second tends to 0 by Theorem 9.3 applied to the integrable function $(g(t)-g(0+))/t$. ■

Theorem 9.6 (9.2.4, Dini's test)

Let $f \in \mathcal{R}^1[-\pi,\pi]$ and $x \in [-\pi,\pi]$. Suppose (a) $f(x+)$ and $f(x-)$ exist, and (b) $\int_0^{\delta}\frac{|f(x+t)+f(x-t)-f(x+)-f(x-)|}{t}\,dt < \infty$. Then $s_n(f;x) \to \frac{1}{2}[f(x+)+f(x-)]$.

Proof. Apply Theorem 9.4 and Lemma 9.5 to $g(t) = \frac{1}{2}[f(x+t)+f(x-t)]$, with $\lambda = n+\frac{1}{2}$. ■

Corollary 9.7 (9.2.5)

If f satisfies a Lipschitz condition at x, $|f(x+t)-f(x)| \le Mt$ for small $|t|$, then $s_n(f;x) \to f(x)$.

Proof. The integrand in (b) is at most $2M$. ■

Corollary 9.8 (9.2.6)

If f is differentiable at x, then $s_n(f;x) \to f(x)$.

Proof. Differentiability at x implies $|f(x+t)-f(x)| \le (|f'(x)|+1)|t|$ for small t. ■

Lemma 9.9 (9.2.7)

If $g:[0,\delta]\to\mathbb{R}$ is of bounded variation, then $\lim_{\lambda\to\infty}\frac{2}{\pi}\int_0^\delta g(t)\frac{\sin\lambda t}{t}\,dt = g(0+)$.

Proof. Take g increasing (Jordan) with $g(0+)=0$, and $h\in(0,\delta)$ with $g(h)<\epsilon$. On $[h,\delta]$, $g(t)/t$ is integrable, so that part $\to 0$ (Theorem 9.3). On $(0,h]$, the second mean value theorem (Corollary 5.25 of Volume I, on $[\eta,h]$, $\eta\to 0$) gives $\int_0^h g(t)\frac{\sin\lambda t}{t}dt = g(h-)\int_c^h \frac{\sin\lambda t}{t}dt$, $c\in[0,h]$; as $|\int_a^b \frac{\sin u}{u}du|\le K$ for $0\le a<b$, this is $\le \epsilon K$. ■

Theorem 9.10 (9.2.8, Dirichlet–Jordan)

If $f\in\mathcal{R}^1[-\pi,\pi]$ is of bounded variation on some interval $[x-\delta,x+\delta]$, then $s_n(f;x)\to\frac{1}{2}[f(x+)+f(x-)]$.

Proof. $g(t)=\frac{1}{2}[f(x+t)+f(x-t)]$ is of bounded variation on $[0,\delta]$ with $g(0+)=\frac{1}{2}[f(x+)+f(x-)]$; apply Theorem 9.4 and Lemma 9.9. ■

Example 9.2

For $f(x)=x$ on $(-\pi,\pi)$ the series converges to x inside, to 0 at $\pm\pi$; $x=\pi/2$ gives $\frac{\pi}{4}=1-\frac{1}{3}+\frac{1}{5}-\cdots$. Near the jump partial sums overshoot by about 9% (Gibbs). Some continuous functions have Fourier series divergent at a point (du Bois-Reymond).

9.3 Fejér's theorem

Motivation. Averaged partial sums converge uniformly for every continuous periodic function.

The **Cesàro means** are $\sigma_n(f;x)=\frac{1}{n+1}\sum_{k=0}^n s_k(f;x)=\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)K_n(t)\,dt$ with the **Fejér kernel**

$$K_n(t)=\frac{1}{n+1}\sum_{k=0}^{n}D_k(t)=\frac{1}{n+1}\Big(\frac{\sin\frac{(n+1)t}{2}}{\sin\frac{t}{2}}\Big)^2.$$

It satisfies $K_n\ge 0$, $\frac{1}{2\pi}\int K_n=1$, and $K_n(t)\le\frac{1}{(n+1)\sin^2(\delta/2)}\to 0$ uniformly on $\delta\le|t|\le\pi$. (The formula follows from $\sum_{k\le n}\sin(k+\frac{1}{2})t\cdot\sin\frac{t}{2}=\frac{1}{2}\sum(\cos kt-\cos(k+1)t)=\sin^2\frac{(n+1)t}{2}$.)

Theorem 9.11 (9.3.1, Fejér)

If $f\in\mathcal{R}^1[-\pi,\pi]$ and $f(x\pm)$ exist, then $\sigma_n(f;x)\to\frac{1}{2}[f(x+)+f(x-)]$. If f is continuous and 2π-periodic, $\sigma_n(f)\to f$ uniformly.

Proof. With $s=\frac{1}{2}[f(x+)+f(x-)]$ and K_n even, $\sigma_n(f;x)-s=\frac{1}{2\pi}\int_0^\pi[f(x+t)+f(x-t)-2s]K_n(t)\,dt$. Take δ with $|f(x+t)+f(x-t)-2s|<\epsilon$ on $(0,\delta)$: that part is $\le\epsilon\cdot\frac{1}{2\pi}\int_0^\pi K_n\le\epsilon/2$, and the part over $[\delta,\pi]$ is $\le\frac{1}{(n+1)\sin^2(\delta/2)}\cdot\frac{1}{2\pi}\int|f(x+t)+f(x-t)-2s|\to 0$. For continuous periodic f, δ is uniform in x and f bounded. ■

So continuous periodic functions are uniform limits of trigonometric polynomials (trigonometric Weierstrass) and are determined by their Fourier coefficients.

Lemma 9.12 (9.3.2)

If $s_n \to s$, then $\sigma_n = \frac{1}{n+1}\sum_{k\le n} s_k \to s$.

Proof. We may assume $s = 0$. Given ϵ, choose N with $|s_n| < \epsilon$ for $n \ge N$. For $n > N$, $|\sigma_n| \le \frac{1}{n+1}\sum_{k<N}|s_k| + \epsilon < 2\epsilon$ for large n. ■

Corollary 9.13 (9.3.3)

If the Fourier series of a continuous periodic f converges at x, its sum is $f(x)$.

Proof. By Lemma 9.12 the limit equals $\lim \sigma_n(f;x) = f(x)$ (Theorem 9.11). ■

Corollary 9.14 (9.3.4)

If $\int_{-\pi}^{\pi}|f|^2 < \infty$, then $\lim_n \|f - s_n(f)\|_2 = 0$.

Proof. Take continuous g vanishing near $\pm\pi$ with $\|f-g\|_2 < \epsilon$ (Theorem 7.14, Lemma 7.13) and $T = \sigma_m(g)$ with $\|g - T\|_{\sup} < \epsilon$ (Theorem 9.11), so $\|g - T\|_2 < \epsilon$. For $n \ge m$, $\|f - s_n(f)\|_2 \le \|f - T\|_2 < 2\epsilon$ (Theorem 9.1). ■

Corollary 9.15 (9.3.5, Parseval's identity)

If $I = [a, a+2\pi]$, $f \in \mathcal{R}^2(I)$ and $c_n = \frac{1}{2\pi}\int_I f(t)e^{-int}dt$, then $\frac{1}{2\pi}\int_I |f|^2 = \sum_{n\in\mathbb{Z}}|c_n|^2$; more generally $\frac{1}{2\pi}\int_I f\bar{g} = \sum c_n\overline{d_n}$ for $g \in \mathcal{R}^2(I)$ with coefficients d_n.

Proof. By periodicity the coefficients do not depend on a. From the proof of Theorem 9.1, $\|f\|_2^2 = \|f - s_n(f)\|_2^2 + \sum_{|k|\le n}|c_k|^2$; let $n \to \infty$ (Corollary 9.14). The polarized form follows from $4\langle f, g\rangle = \sum_{k=0}^{3} i^k\|f + i^k g\|^2$. ■

Example 9.3

Parseval for $f(x) = x$ gives $\frac{1}{2\pi}\int_{-\pi}^{\pi} x^2 dx = \frac{\pi^2}{3} = 2\sum_{n\ge1}\frac{1}{n^2}$, again $\sum n^{-2} = \pi^2/6$.

9.4 Differentiation and integration of Fourier series

Motivation. Fourier series integrate termwise always, and differentiate termwise under smoothness.

Theorem 9.16 (9.4.1, termwise integration)

If $f \in \mathcal{R}^2(I)$ has Fourier series $\sum \hat{f}(n)e^{int}$, then for $a, x \in I$, $\int_a^x f(t)\,dt = \sum_n \hat{f}(n)\int_a^x e^{int}dt$, the series converging (uniformly in x).

Proof. By Cauchy–Schwarz, $\left|\int_a^x (f - s_n(f))\right| \le \sqrt{2\pi}\left(\int_I |f - s_n(f)|^2\right)^{1/2} \to 0$ by Corollary 9.14. ■

This holds even if the Fourier series of f diverges. For periodic C^1 f, $\widehat{f'}(n) = in\hat{f}(n)$; $\sum |n\hat{f}(n)|^2 < \infty$ and Cauchy–Schwarz give $\sum|\hat{f}(n)| < \infty$, so the series converges uniformly, to f (Corollary 9.13).

A sequence $a : \mathbb{Z} \to \mathbb{C}$ is **rapidly decreasing** if $|n|^k a(n)$ is bounded for every k.

Theorem 9.17 (9.4.2)

The Fourier coefficients of a C^∞ periodic function are rapidly decreasing. Conversely, if a is rapidly decreasing, $f(x) = \lim_n \sum_{|k|\le n} a(k)e^{ikx}$ defines a C^∞ periodic function with $\hat{f}(n) = a(n)$.

Proof. $\widehat{f^{(k)}}(n) = (in)^k \hat{f}(n)$ (by parts), so $|n^k \hat{f}(n)| \le \frac{1}{2\pi}\int |f^{(k)}|$. Conversely all termwise derivatives $\sum (ik)^j a(k)e^{ikx}$ converge uniformly (M-test, $\sum |k|^j |a(k)| < \infty$), so f is C^∞ (Theorem 6.4), and termwise integration against e^{-inx} gives $\hat{f}(n) = a(n)$. ∎

Lemma 9.18 (9.4.3)

If $f : (a, b) \to \mathbb{R}$ is analytic and $K \subset (a, b)$ is compact, there are $C, R > 0$ with $|f^{(n)}(x)| \le Cn!R^n$ for $x \in K$, $n = 1, 2, \ldots$.

Proof. Fix $c \in K$; near c, $f(x) = \sum_k a_k(x - c)^k$ with positive radius, so there are $\rho > 0$, M with $|a_k|\rho^k \le M$. For $|x - c| \le \rho/2$, termwise differentiation (Theorem 6.12) gives

$$|f^{(n)}(x)| \le \sum_{k\ge n} \frac{k!}{(k-n)!}|a_k|\Big(\frac{\rho}{2}\Big)^{k-n} \le M\rho^{-n}n! \sum_{j\ge 0} \binom{n+j}{n} 2^{-j} = 2M\, n!\Big(\frac{2}{\rho}\Big)^n.$$

Finitely many such neighbourhoods cover K; take the largest constants. ∎

Lemma 9.19 (9.4.4)

If $c \ge 0$, $R > 0$, $n \ge 1$ and $c \le k!\,(R/n)^k$ for all $k = 0, 1, 2, \ldots$, then $c \le (R + 1)e^{-n/(R+1)}$.

Proof. If $c > 0$, then $\frac{1}{k!}\big(\frac{n}{R+1}\big)^k \le \frac{1}{c}\big(\frac{R}{R+1}\big)^k$ for every k; summing, $e^{n/(R+1)} \le \frac{R+1}{c}$. ∎

Theorem 9.20 (9.4.5)

If $f : \mathbb{R} \to \mathbb{C}$ is periodic and analytic, then $\limsup_{|n|\to\infty} |\hat{f}(n)|^{1/|n|} < 1$.

Proof. By Lemma 9.18 on $K = [-\pi, \pi]$, $\|f^{(k)}\|_{\sup} \le Ck!R^k$ for $k \ge 1$ (and enlarging C, also for $k = 0$). Then $|\hat{f}(n)| = |\widehat{f^{(k)}}(n)|/|n|^k \le Ck!(R/|n|)^k$ for all k, so by Lemma 9.19, $|\hat{f}(n)| \le C(R + 1)e^{-|n|/(R+1)}$, and $\limsup |\hat{f}(n)|^{1/|n|} \le e^{-1/(R+1)} < 1$. ∎

Conversely, such geometric decay of the coefficients makes f analytic (the series extends to a holomorphic function near the unit circle).

Theorem 9.21 (9.4.6)

If $\int_{-\pi}^{\pi} |f| < \infty$, then $\lim_{|n|\to\infty} \hat{f}(n) = 0$.

Proof. Theorem 9.3 applied to the real and imaginary parts of $\int f(t)(\cos nt - i \sin nt)\, dt$. ∎

The decay can be arbitrarily slow, but not every sequence tending to 0 is a sequence of Fourier coefficients:

Proposition 9.22 (9.4.7)

If $f \in \mathcal{R}^1[-\pi,\pi]$ is odd, then the partial sums of $\sum_{n\ge1}\hat f(n)/n$ are bounded.

Proof. For odd f, $\sum_{n\le N}\frac{\hat f(n)}{n} = -\frac{i}{\pi}\int_0^\pi f(t)\sum_{n\le N}\frac{\sin nt}{n}\,dt$. On $(0,\pi]$, $|\sum_{n\le N}\frac{\sin nt}{n}| \le 1+2\pi$: terms with $n \le 1/t$ are $\le t$ each, and summation by parts with (‡) of Chapter 6 bounds the rest by $2\cdot t\cdot\pi/t$. So the partial sums are $\le \frac{1+2\pi}{\pi}\int_0^\pi|f|$. ∎

So $\sum_{n\ge2}\frac{\sin nx}{\log n}$, which converges everywhere (Theorem 6.20), is not the Fourier series of any $f\in\mathcal{R}^1$, since $\sum\frac{1}{n\log n}=\infty$.

9.5 Selected exercises

Motivation. Book exercises: Lebesgue constants, regular averaging, and $1/|n|$ decay under bounded variation.

Exercise 9.5.3 (Lebesgue constants)

Show that $\frac{1}{2\pi}\int_{-\pi}^{\pi}|D_n(t)|\,dt\to\infty$ as $n\to\infty$; in fact it is at least $\frac{4}{\pi^2}(1+\frac12+\cdots+\frac1n)$.

Solution.

Step 1. D_n is even and $0<\sin\frac{t}{2}\le\frac{t}{2}$ on $(0,\pi]$, so

$$\frac{1}{2\pi}\int_{-\pi}^{\pi}|D_n| = \frac{1}{\pi}\int_0^\pi\frac{|\sin(n+\frac12)t|}{\sin\frac{t}{2}}\,dt \ge \frac{2}{\pi}\int_0^\pi\frac{|\sin(n+\frac12)t|}{t}\,dt = \frac{2}{\pi}\int_0^{(n+\frac12)\pi}\frac{|\sin u|}{u}\,du,$$

by the substitution $u=(n+\frac12)t$.

Step 2. On $[k\pi,(k+1)\pi]$, $\frac1u\ge\frac{1}{(k+1)\pi}$ and $\int_{k\pi}^{(k+1)\pi}|\sin u|\,du=2$. Summing over $k=0,\dots,n-1$,

$$\frac{1}{2\pi}\int_{-\pi}^{\pi}|D_n| \ge \frac{2}{\pi}\sum_{k=0}^{n-1}\frac{2}{(k+1)\pi} = \frac{4}{\pi^2}\sum_{k=1}^{n}\frac1k\to\infty.$$

By contrast $\frac{1}{2\pi}\int|K_n|=1$ ($K_n\ge0$): hence Fejér's theorem holds for all continuous f while partial sums may diverge (Example 9.2). ∎

Exercise 9.5.8 (Regular averaging methods)

Let $a_{mn}\ge0$ $(m,n\ge1)$ with $a_{mn}=0$ for $n>m$ and $\sum_{n=1}^m a_{mn}=1$ for every m. For a sequence (s_n) put $t_m=\sum_{n=1}^m a_{mn}s_n$. Show that the following are equivalent:
(a) for every convergent sequence, $\lim_n s_n=s$ implies $\lim_m t_m=s$;
(b) $\lim_{m\to\infty}a_{mn}=0$ for each n.

Solution.

(a)⇒(b): $s_n=\delta_{nn_0}\to0$ gives $t_m=a_{mn_0}\to0$.

(b)⇒(a):

Step 1. $t_m-s=\sum_{n\le m}a_{mn}(s_n-s)$, as $\sum_n a_{mn}=1$.

Step 2. If $|s_n-s|<\epsilon$ for $n>N$, then for $m>N$ $(a_{mn}\ge0)$

$$|t_m-s|\le\sum_{n\le N}a_{mn}|s_n-s|+\epsilon\sum_{N<n\le m}a_{mn}\le\sum_{n\le N}a_{mn}|s_n-s|+\epsilon.$$

Step 3. The first sum has N fixed terms, each $\to 0$ by (b); so $\limsup_m |t_m - s| \le \epsilon$.

Taking $a_{mn} = \frac{1}{m}$ for $n \le m$ recovers Lemma 9.12. ■

Exercise 9.5.13 (Fourier coefficients of functions of bounded variation)

A function $f : [-\pi, \pi] \to \mathbb{C}$ is of bounded variation if $\sup_P \sum_i |f(x_i) - f(x_{i-1})| < \infty$ over all partitions P.
(a) Show that a function of bounded variation is a linear combination of increasing real functions.
(b) Show that for every increasing $f : [-\pi, \pi] \to \mathbb{R}$, $|\hat{f}(n)| \le \frac{V_{-\pi}^{\pi}(f)}{\pi |n|}$ for $n = \pm 1, \pm 2, \dots$.
(c) Show that the Fourier coefficients of a function f of bounded variation satisfy $|\hat{f}(n)| \le \frac{C}{|n|}$ $(n \neq 0)$ for some constant $C > 0$.

Solution.

(a) $\operatorname{Re} f, \operatorname{Im} f$ are of bounded variation ($|\operatorname{Re} z|, |\operatorname{Im} z| \le |z|$); Jordan (Theorem 5.16 of Volume I) gives $\operatorname{Re} f = g_1 - h_1$, $\operatorname{Im} f = g_2 - h_2$ with g_j, h_j increasing.

(b) Let $V = V_{-\pi}^{\pi}(f) = f(\pi) - f(-\pi)$ and $n \neq 0$.

Step 1. Choose $\theta \in \mathbb{R}$ with $e^{i\theta} \int_{-\pi}^{\pi} f(t) e^{-int} dt \ge 0$. Then

$$2\pi |\hat{f}(n)| = \operatorname{Re}\Big(e^{i\theta} \int_{-\pi}^{\pi} f(t) e^{-int} dt \Big) = \int_{-\pi}^{\pi} f(t) \cos(nt - \theta)\, dt.$$

Step 2. By the second mean value theorem (Corollary 5.25 of Volume I) there is $c \in [-\pi, \pi]$ with

$$\int_{-\pi}^{\pi} f(t) \cos(nt - \theta)\, dt = f(-\pi) \int_{-\pi}^{c} \cos(nt - \theta)\, dt + f(\pi) \int_{c}^{\pi} \cos(nt - \theta)\, dt.$$

Step 3. As $\int_{-\pi}^{\pi} \cos(nt - \theta)\, dt = 0$, the right side is $-V \int_{-\pi}^{c} \cos(nt - \theta)\, dt = -\frac{V}{n} (\sin(nc - \theta) - \sin(-n\pi - \theta))$, of absolute value $\le \frac{2V}{|n|}$.

Step 4. So $2\pi |\hat{f}(n)| \le \frac{2V}{|n|}$.

(c) With $f = g_1 - h_1 + i g_2 - i h_2$ from (a), linearity and (b) give

$$|\hat{f}(n)| \le |\hat{g}_1(n)| + |\hat{h}_1(n)| + |\hat{g}_2(n)| + |\hat{h}_2(n)| \le \frac{C}{|n|},$$

with $C = \frac{1}{\pi}(V(g_1) + V(h_1) + V(g_2) + V(h_2))$, $V = V_{-\pi}^{\pi}$. $f(x) = x$ ($|\hat{f}(n)| = \frac{1}{|n|}$, Example 9.1) shows the order is sharp. ■